

Alcohol, Phenols and Ethers

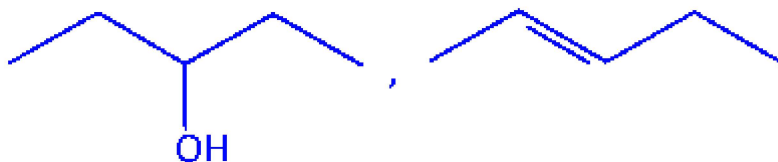
Question1

An alcohol X ($C_5H_{12}O$) on dehydration gives Y (major product). Reaction of Y with HBr gave Z ($C_5H_{11}Br$), major product). Z undergoes nucleophilic substitution in two steps. What are X and Y ?

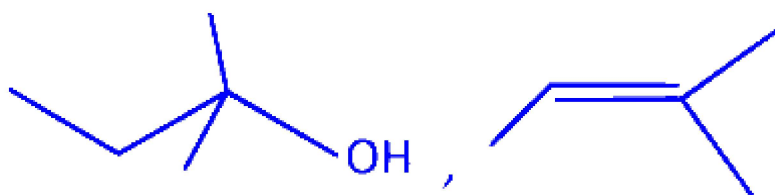
AP EAPCET 2025 - 26th May Morning Shift

Options:

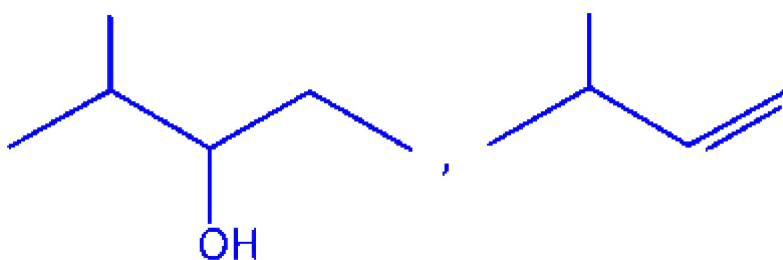
A.



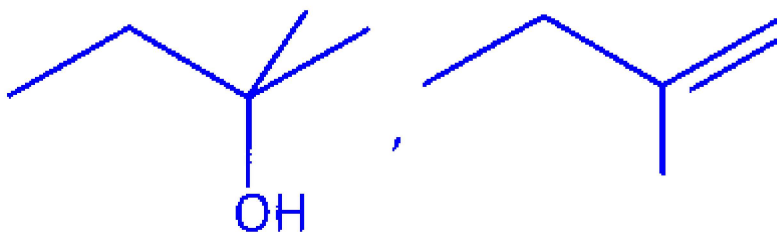
B.



C.



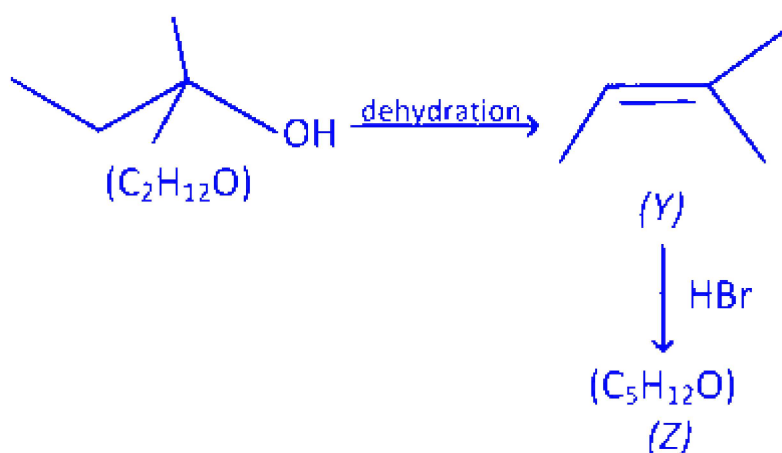
D.



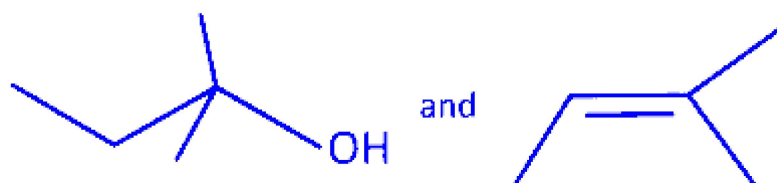
Answer: B

Solution:

The complete reaction sequence is as follows

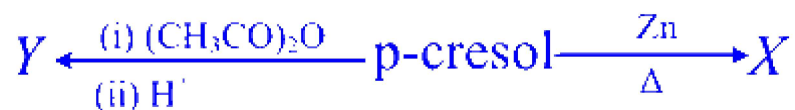


Thus the compound X and Y are



Question2

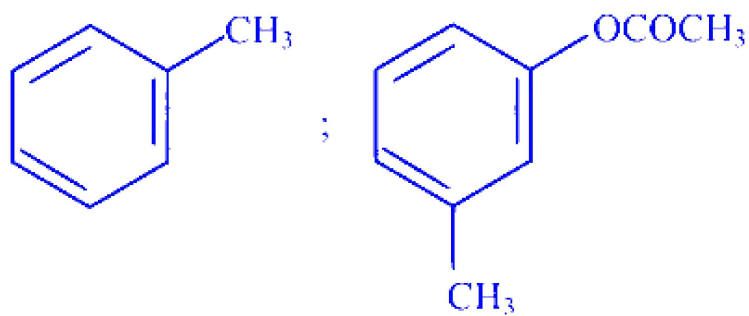
What are X and Y respectively in the following set of reactions?



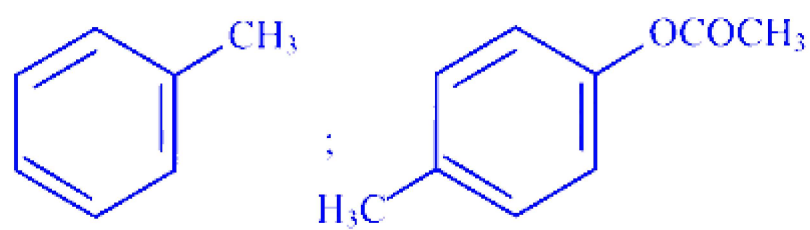
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Options:

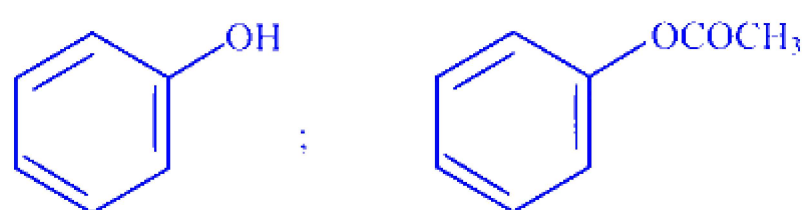
A.



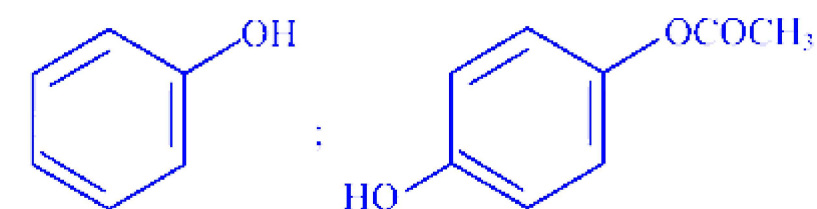
B.



C.



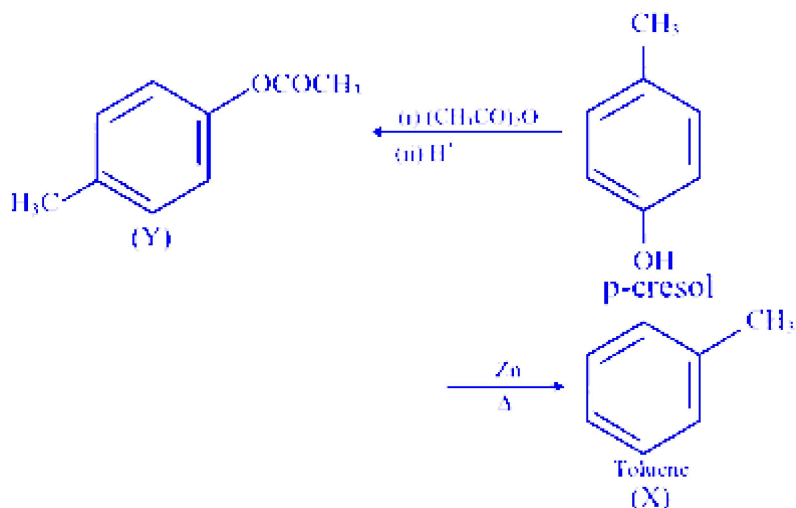
D.



Answer: B

Solution:

The complete reaction sequence is as follows



Question3

Match the following

List-I (Compound)		List-II (pK_a)	
(A)	p-nitrophenol	(I)	15.9
(B)	Phenol	(II)	7.1
(C)	Ethanol	(III)	10.0
(D)	p-cresol	(IV)	10.2
		(N)	8.3

The correct answer is

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Options:

A.

A-II, B-V, C-I, D-III

B.

A-II, B-III, C-I, D-IV

C.

A-V, B-IV, C-II, D-III

D.

A-IV, B-III, C-I, D-V

Answer: B

Solution:

1. **Understand pKa and Acidity:** A lower pKa value indicates a stronger acid.

2. **Analyze each compound:**

- **(C) Ethanol:** Alcohols are generally weak acids. Their acidity is similar to water. The pKa of ethanol is known to be around 16.
 - Looking at List-II, (I) 15.9 is a perfect match for Ethanol.
 - So, **C-I**.
- **(B) Phenol:** Phenol is significantly more acidic than alcohols due to the resonance stabilization of its conjugate base (phenoxide ion). Its pKa is typically around 10.
 - Looking at List-II, (III) 10.0 is a direct match for Phenol.
 - So, **B-III**.
- **(A) p-nitrophenol:** The nitro group (-NO₂) is a strong electron-withdrawing group (EWG) by both inductive (-I) and resonance (-R) effects. When an EWG is at the para position, it strongly stabilizes the phenoxide ion through resonance, making p-nitrophenol much more acidic than phenol. Therefore, its pKa should be significantly lower than 10.
 - Looking at List-II, (II) 7.1 and (V) 8.3 are options below 10.0. The nitro group typically lowers the pKa of phenol by about 2-3 units. So, $10.0 - 2.9 = 7.1$.
 - Thus, (II) 7.1 is an excellent match for p-nitrophenol.
 - So, **A-II**.
- **(D) p-cresol:** p-cresol is 4-methylphenol. The methyl group (-CH₃) is an electron-donating group (EDG) due to hyperconjugation and inductive effect (+I). EDGs destabilize the phenoxide ion, making p-cresol slightly less acidic than phenol. Therefore, its pKa should be slightly higher than phenol's pKa (10.0).
 - Looking at List-II, we have (IV) 10.2 left. This value is slightly higher than 10.0 (phenol), which is consistent with the presence of an EDG.
 - So, **D-IV**.

3. **Consolidate the matches:**

- (A) p-nitrophenol - (II) 7.1
- (B) Phenol - (III) 10.0
- (C) Ethanol - (I) 15.9

- (D) p-cresol - (IV) 10.2

4. Check the options:

- Option A: A-II, B-V, C-I, D-III (Incorrect, B is 8.3, D is 10.0)
- **Option B: A-II, B-III, C-I, D-IV (This perfectly matches our derived pairs.)**
- Option C: A-V, B-IV, C-II, D-III (Incorrect, C is 7.1)
- Option D: A-IV, B-III, C-I, D-V (Incorrect, A is 10.2, D is 8.3)

Therefore, Option B is the correct answer.

The final answer is B

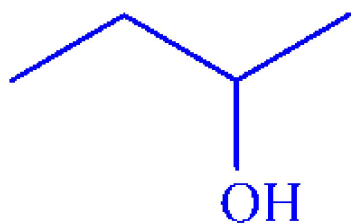
Question4

An alcohol X ($C_4H_{10}O$) on dehydration gave alkene (C_4H_8) as major product, which on bromination followed by treatment with Y gave alkyne C_4H_6 . Alkyne C_4H_6 , does not react with sodium metal. What are X and Y ?

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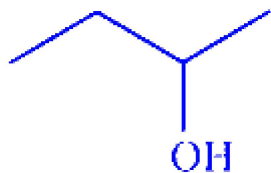
Options:

A.



; alc. KOH

B.



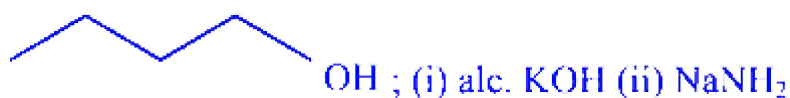
; (i) alc. KOH (ii) $NaNH_2$

C.



OH ; alc. KOH

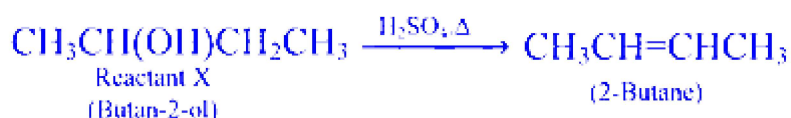
D.



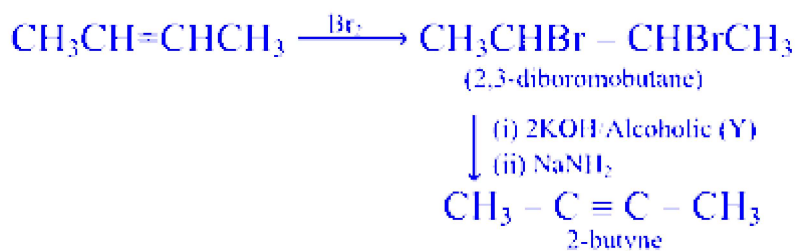
Answer: B

Solution:

The complete mechanism of reaction is,



Bromination of alkene



X = Butan-2-ol, Y = KOH

Question5

The correct statement regarding X and Y formed in the following reaction is



AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

X undergoes substitution by S_N2 mechanism

B.

X undergoes substitution with water in two steps

C.

Y gets converted to corresponding chloride with conc. HCl at room temperature

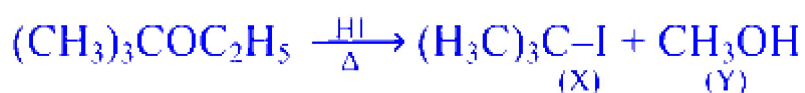
D.

Reaction of Y with Cu at 573 K gives ketone

Answer: B

Solution:

The complete reaction involved is



Thus, X undergoes substitution with water in two steps.

Question6

The correct order of boiling points of the compounds given below is

(A) methoxy ethane

(B) propan-1-ol

(C) propanal

(D) propanone

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Options:

A.

$\text{C} > \text{B} > \text{A} > \text{D}$

B.

$B > D > C > A$

C.

$B > C > D > A$

D.

$C > A > B > D$

Answer: B

Solution:

The correct order of boiling point is,

$B > D > C > A$

Alcohol has highest boiling point due to hydrogen bonding.

Order is,

Alcohol > Ketone > Aldehyde > Ether

Question7

The number of primary (1°), secondary (2°) and tertiary (3°) alcohols possible for the formula $C_5H_{12}O$ respectively are

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

3, 3, 2

B.

4, 2, 2

C.

4, 3, 1

D.

3, 4, 1

Answer: C

Solution:

There are 4 primary alcohol for $C_5H_{12}O$ they are pentan-1-ol, 2-methylbutan-1-ol, 3-methylbutan-1-ol and 2,2-dimethylpropan-1-ol. There are 3 secondary alcohol possible for $C_5H_{12}O$ which are pentan-2-ol, pentan-3-ol and methylbutan-2-ol. Only one tertiary alcohol possible for $C_5H_{12}O$ which is 2-methylbutan-2-ol.

Question8

In which of the following intramolecular H -bonding is absent?

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

Salicylic acid

B.

Salicylaldehyde

C.

Quinol

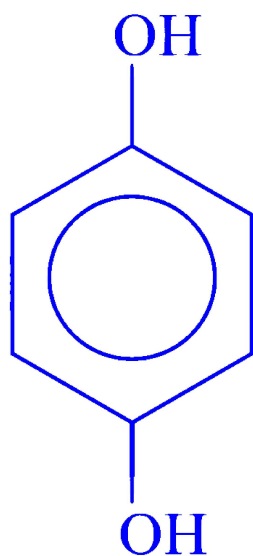
D.

Catechol

Answer: C

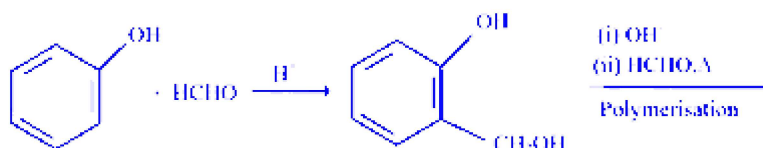
Solution:

In quinol, intramolecular hydrogen bonding is absent



Question9

The *X* formed in the following reaction sequence and its structural type are respectively.



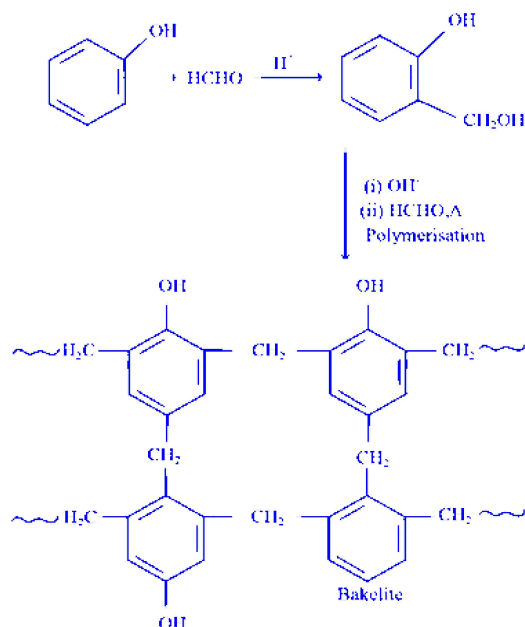
AP EAPCET 2024 - 23th May Morning Shift

Options:

- A. Novolac - linear polymer
- B. Bakelite - cross linked polymer
- C. Novolac - cross linked polymer
- D. Bakelite - linear polymer

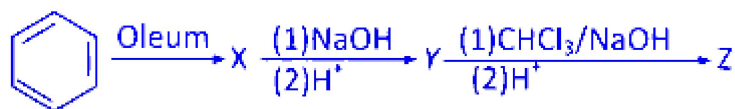
Answer: B

Solution:



Question10

What is the major product ' Z ' in the following reaction sequence?



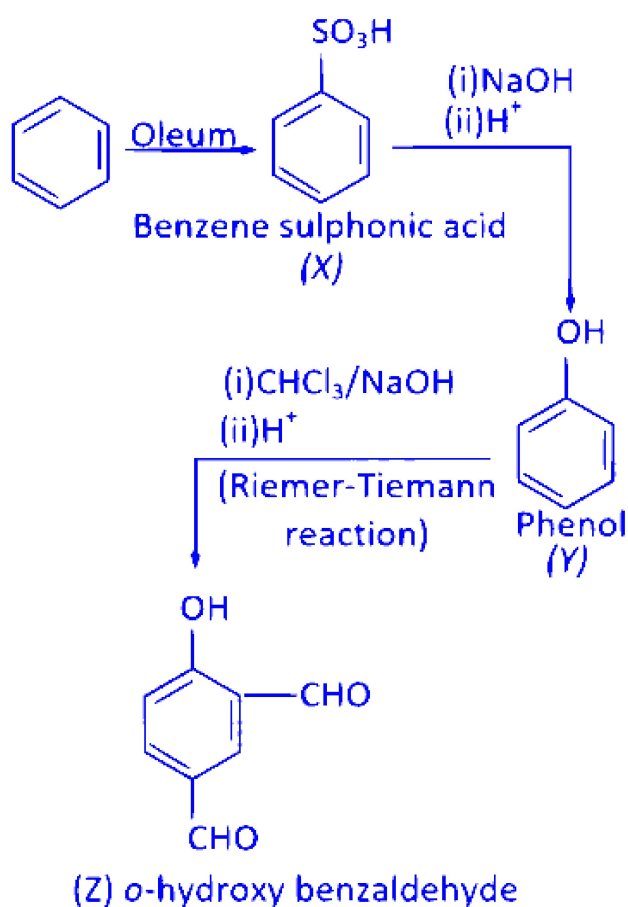
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Options:

- A. o- hydroxy benzaldehyde
- B. p- hydroxy benzaldehyde
- C. o - hydroxy benzoic acid
- D. p- hydroxy benzoic acid

Answer: A

Solution:

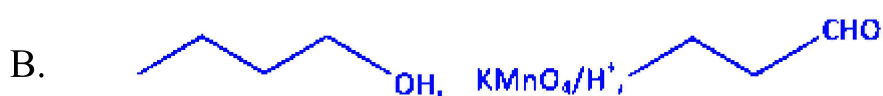
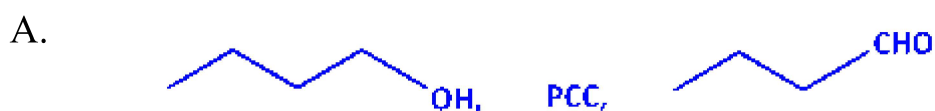


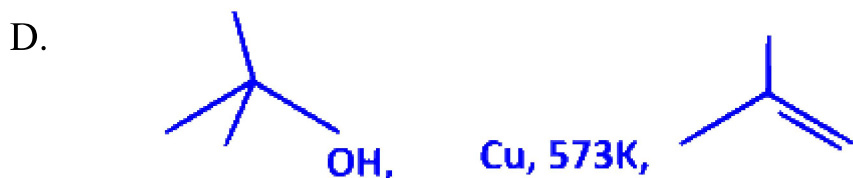
Question11

An alcohol X ($\text{C}_4\text{H}_{10}\text{O}$) does not give turbidity with conc. HCl and ZnCl_2 at room temperature. X on reaction with reagent Y gives Z . What are X , Y and Z respectively?

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Options:

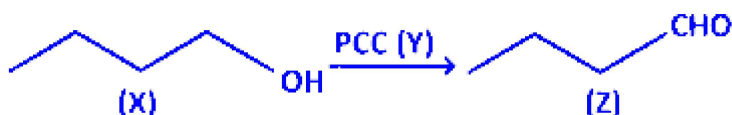




Answer: A

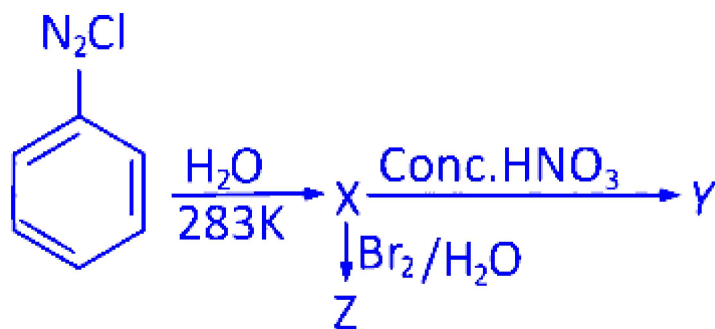
Solution:

Since the alcohol X does not give turbidity with conc. HCl and ZnCl_2 at room temperature. Thus, the X is primary alcohol. The required reaction is as follows.



Question12

Consider the following reactions,



Y and Z respectively are

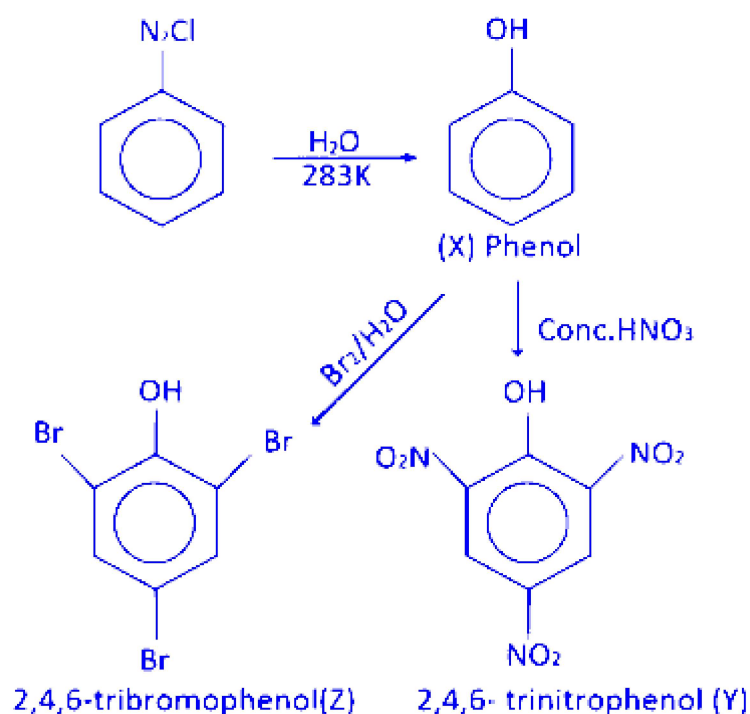
AP EAPCET 2024 - 20th May Morning Shift

Options:

- A. Picric acid (2, 4, 6-trinitrophenol, 2, 4, 6-tribromophend
- B. o-nitrophenol, *p*-bromophenol
- C. *p*-nitrophenol o-bromophenol
- D. 2, 4-dinitrophenol, 2-4 dibromophenol

Answer: A

Solution:



Thus, the compounds Y and Z are 2, 4, 6-trinitrophenol (picric acid) and 2, 4, 6-tribromophenol.

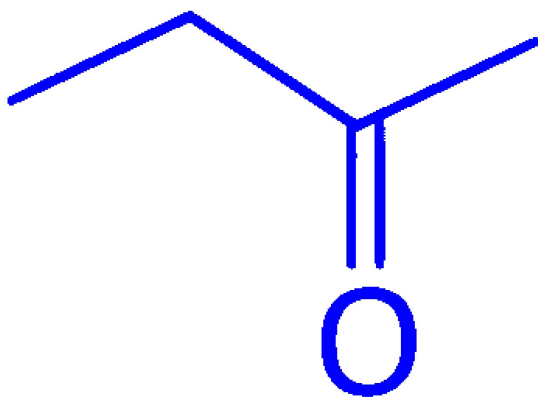
Question13

An alcohol X ($\text{C}_4\text{H}_{10}\text{O}$) reacts with conc. HCl at room temperature to give Y ($\text{C}_4\text{H}_9\text{Cl}$). Reaction of X with copper at 573 K gave Z. What is Z ?

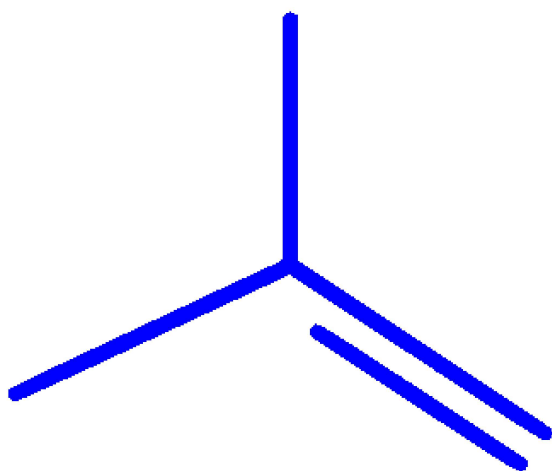
AP EAPCET 2024 - 19th May Evening Shift

Options:

A.



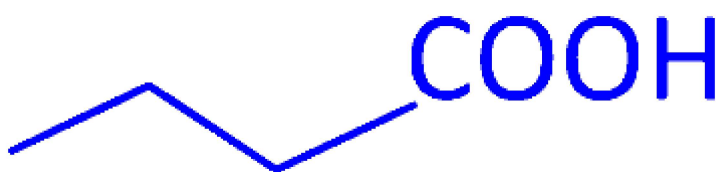
B.



C.

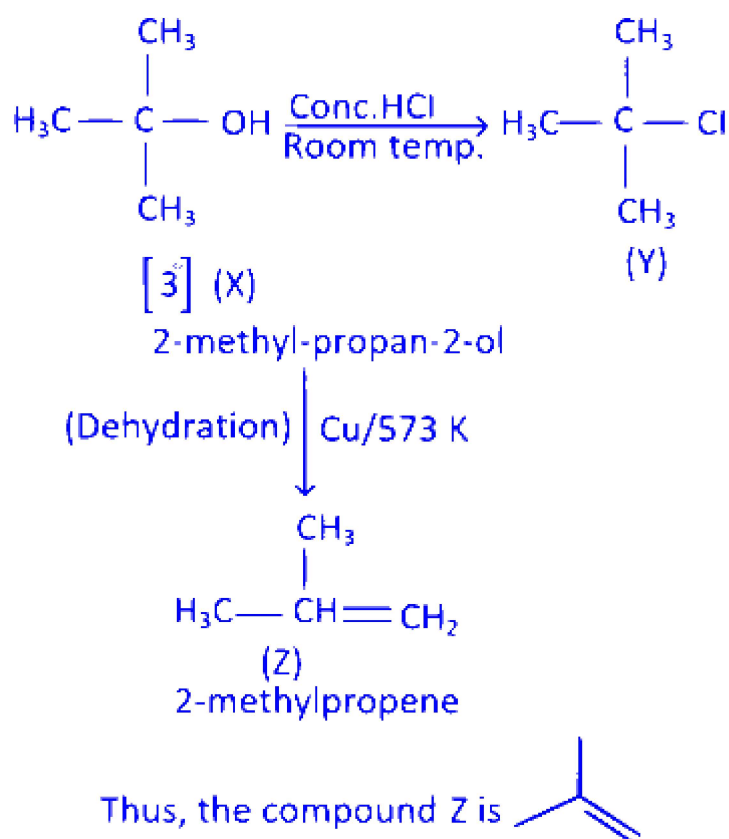


D.



Answer: B

Solution:

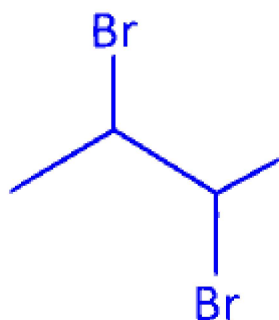


A dibromide X ($\text{C}_4\text{H}_4\text{Br}_2$) on dehydrohalogenation gave Y which on reduction with Z gave non-polar isomer of C_4H_{10} . What are X and Z respectively?

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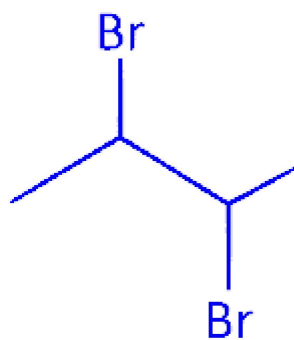
Options:

A.



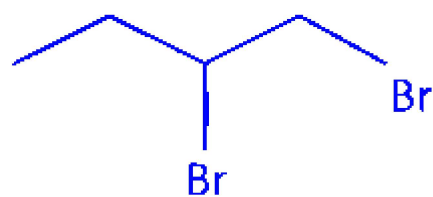
$\text{Na}/\text{NH}_3(\text{l})$

B.



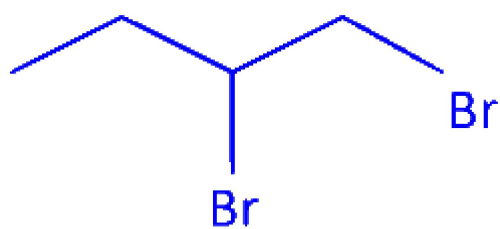
Pd/C

C.



Na/NH₃(l)

D.

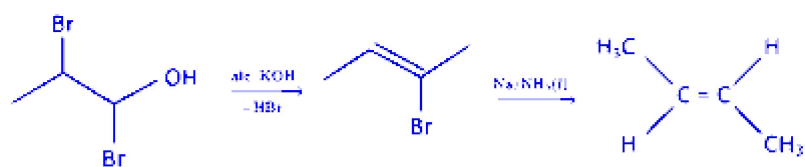


Pd/C

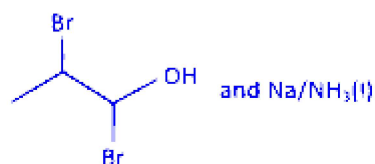
Answer: A

Solution:

The reaction involved is as follows.

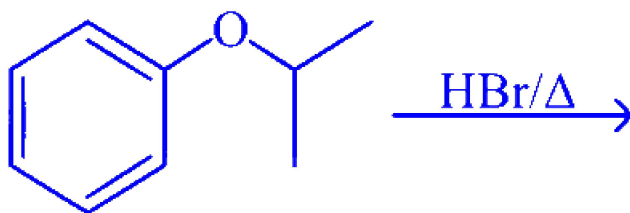


Thus, compound X and Z are



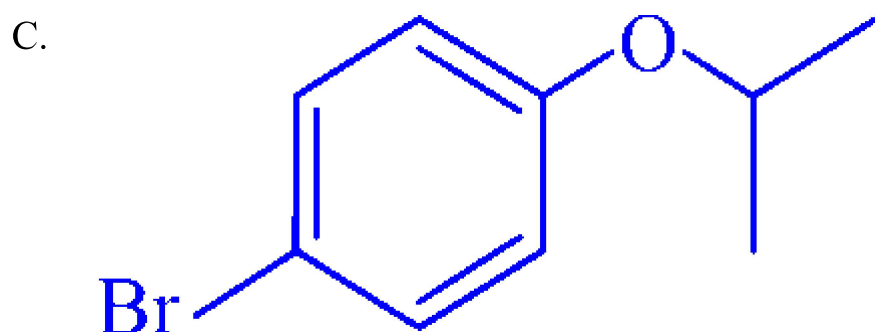
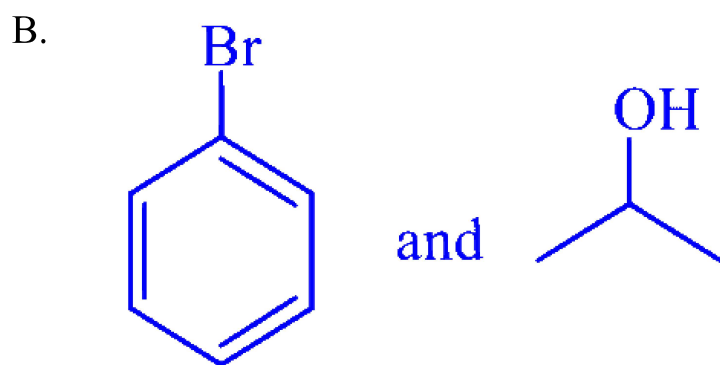
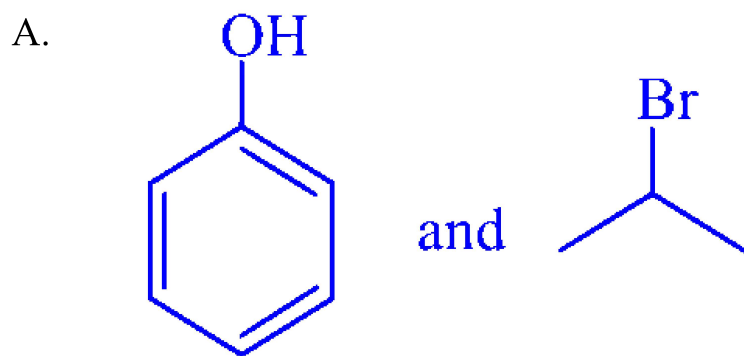
Question15

Identify the product(s) formed in the following reaction.

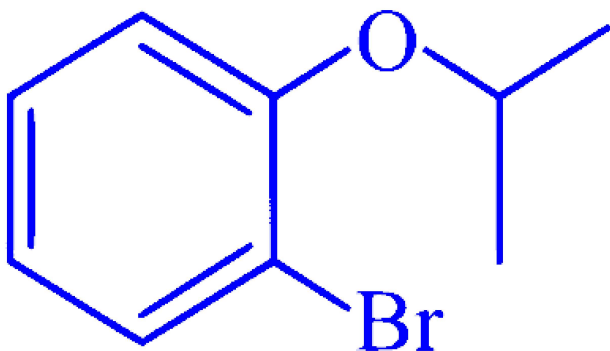


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Options:

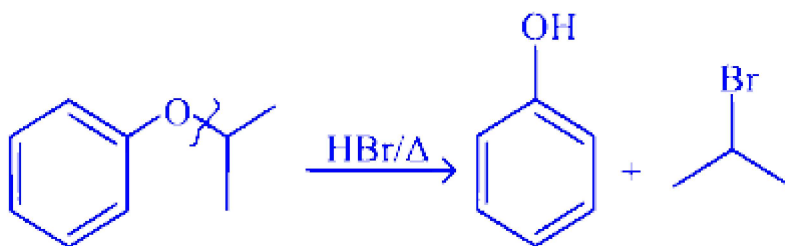


D.



Answer: A

Solution:



Question16

Which compound is formed on catalytic hydrogenation of carbon monoxide at high p and high T in presence of $\text{ZnO} - \text{Cr}_2\text{O}_3$ catalyst?

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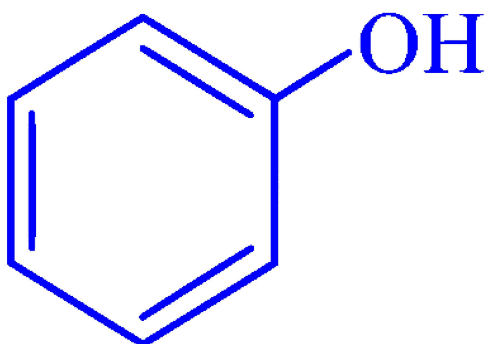
Options:

A. CH_3OH

B. CH_3COOH

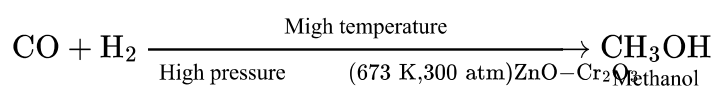
C. $\text{CH}_3\text{CH}_2\text{OH}$

D.



Answer: A

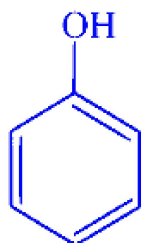
Solution:



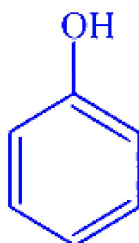
Methanol is industrially prepared when mixture of carbon monoxide and hydrogen is subjected to high temperature and pressure in presence of $\text{ZnO} - \text{Cr}_2\text{O}_3$ as catalyst.

Question17

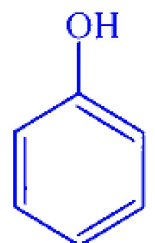
The correct order of acidity of the following compounds is



I



II



III

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. III > II > I

B. II > III > I

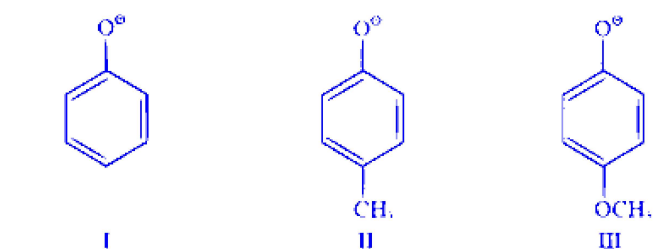
C. I > II > III

D. III > I > II

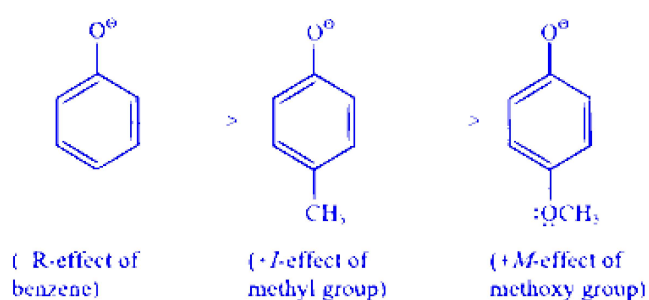
Answer: C

Solution:

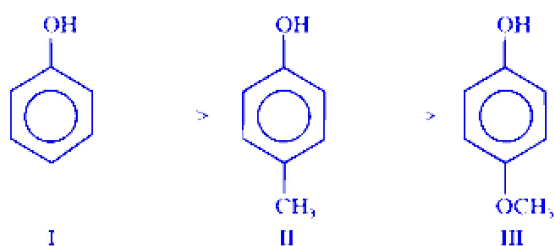
Acidity $\propto$ stability of the anion formed. The anions formed are



Stability order is

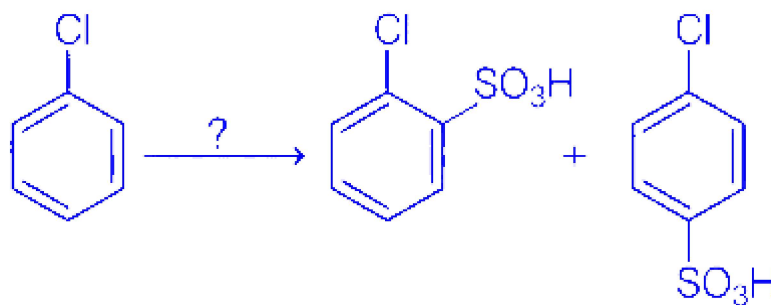


Thus, acidity order is



Question18

Identify the best suitable reagent for the following reaction.



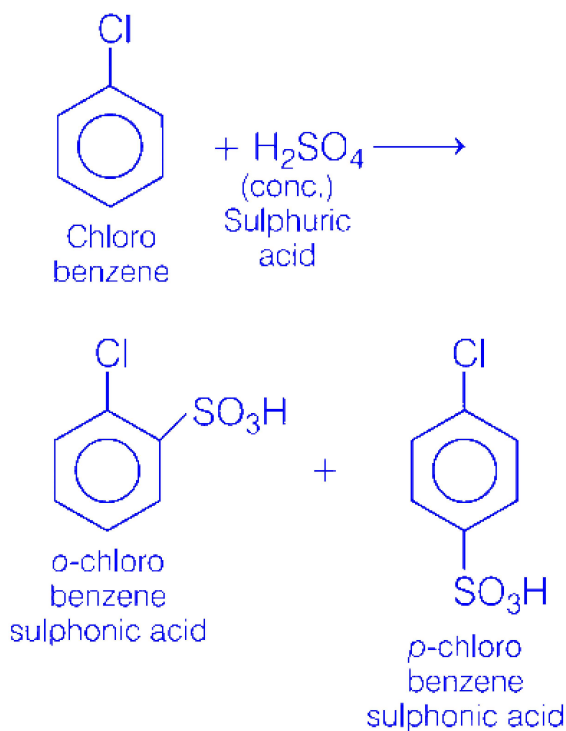
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Options:

- A. Concentrated sulphuric acid
- B. Dilute sulphuric acid (10%)
- C. Concentrated sulphuric acid and concentrated nitric acid
- D. concentrated acetic acid (fuming)

Answer: A

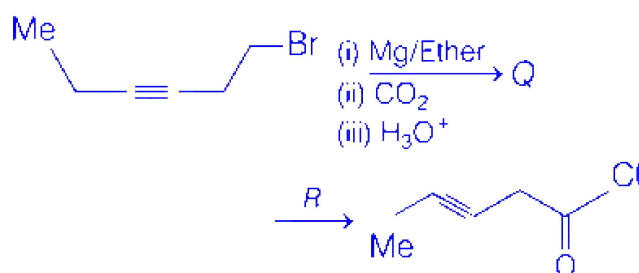
Solution:



Chlorine is an ortho-para directing group. Chlorine is electronegative species but it has lone pair of electrons which has tendency to add in resonance structure of benzene ring which develop negative charge on ortho and para position and minimum electron density at meta position.

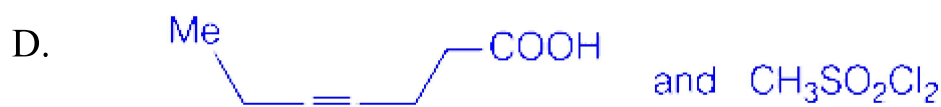
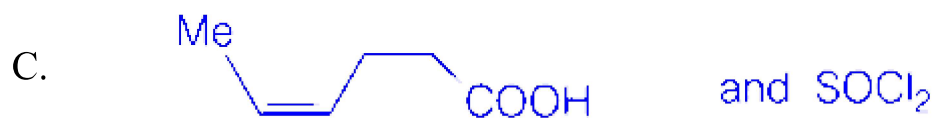
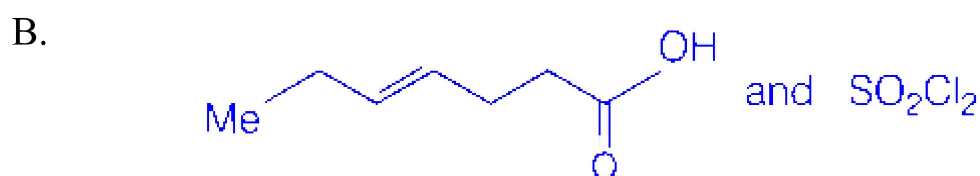
Question19

In the following reaction sequence, identify product 'Q' and reagent 'R'.



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Options:



Answer: A

Solution:

Alkyl bromide in presence of Mg/ether gives RMg^+Br^- . Here, alkyl group is nucleophilic and attacks CO_2 to form carboxylic acid.

$SOCl_2$ is used to replace $-OH$ group to $-Cl$ group and evolved SO_2 and HCl gas.

